6.4 Multiplying and dividing: answers

← Unit overview · Practice

For parents: there are many good ways to split a number. If his split is different from the one here but correct, that's great. Say so.

Set 1 · Warm-up

#	Answer	How
1	30	3 tens.
2	400	4 hundreds.
3	60	6 tens.
4	100	Two fifties.
5	100	Five twenties.
6	450	45 tens.
7	10	Half of 20.
8	10	6 × 10 = 60.
9	10	10 × 10 = 100.
10	60	3 × 2 tens = 6 tens.
11	40	4 tens.

Set 2 · Tens and hundreds

#	Answer	How
1	2 100	3 × 7 hundreds = 21 hundreds.
2	1 200	3 × 4 = 12; tens × tens = hundreds, so 12 hundreds.
3	240	4 × 6 tens = 24 tens.
4	1 800	6 × 3 hundreds = 18 hundreds.
5	1 000	2 × 5 = 10; tens × tens = hundreds, so 10 hundreds.
6	560	8 × 7 tens = 56 tens.
7	280	4 tens × 7 = 28 tens.
8	1 800	9 × 2 hundreds = 18 hundreds.
9	300	5 tens × 6 = 30 tens.
10	100	10 tens.

Set 3 · Splitting

#	Answer	How
1	268	240 + 28.
2	126	3 × 40 + 3 × 2 = 120 + 6.
3	190	150 + 40. Or 5 × 40 − 5 × 2 = 200 − 10.
4	144	120 + 24.

#	Answer	How
5	371	350 + 21.
6	200	160 + 40. Or: 4 × 25 = 100, so 8 × 25 = 200.
7	€84	6 × 14 = 60 + 24.
8	216 bottles	9 × 24 = 180 + 36. Or 10 × 24 − 24 = 240 − 24.
9	86	80 + 6.

Talk about it: yes: 5 × 40 = 200, then take away 5 × 2 = 10, giving 190. Taking away the 2 extra in each of the 5 groups.

Set 4 · Column by column

#	Answer	Working
1	1 704	1 200 + 480 + 24 (estimate 6 × 300 = 1 800)
2	645	600 + 30 + 15
3	652	400 + 240 + 12
4	1 635	1 500 + 100 + 35
5	2 856	2 800 + 0 + 56. The 0 tens give 0.
6	750 pencils	6 × 125 = 600 + 120 + 30
7	€92	4 × 23 = 80 + 12
8	222	200 + 20 + 2
9	2 144	Right to left: 4 × 6 = 24, write 4 carry 2; 4 × 3 = 12, + 2 = 14, write 4 carry 1; 4 × 5 = 20, + 1 = 21.

Set 5 · Dividing

#	Answer	How
1	23	80 : 4 = 20 and 12 : 4 = 3.
2	15	50 : 5 = 10 and 25 : 5 = 5.
3	14	60 : 6 = 10 and 24 : 6 = 4.
4	50	How many 30s in 1 500? Same as 150 : 3 = 50.
5	40	Same as 240 : 6 = 40.
6	9 rest 2	9 × 5 = 45, 47 − 45 = 2.
7	14 rest 2	14 × 7 = 98, 100 − 98 = 2.
8	91 rest 7	9 × 90 = 810, 16 left; 9 × 1 = 9, 7 left. Check: 91 × 9 + 7 = 826.

#	Answer	How
9	15 cars	58 : 4 = 14 rest 2. The 2 children still need a car, so round **up**.
10	12 full boxes	75 : 6 = 12 rest 3. 3 eggs don’t fill a box, so round **down**.
11	100	500 : 5 = 100, and 498 is very close to 500. (Exact: 99 rest 3.)
12	10	5 × 10 = 50.

Puzzle corner

- **Leaves 1 over for 2 and 3:** 7, 13, 19, 25, 31… (one more than a number in both the 2 and 3 tables: 6, 12, 18…).
- **Smallest for 2, 3, 4 and 5: 61 eggs.** 60 is the smallest number in the 2, 3, 4 and 5 tables, so 60 + 1 = 61.
- **And 6?** Yes: 61 : 6 = 10 rest 1.

Finding 7 or 13 is a real success. The jump to 61 is the “big challenge”; a helpful hint is “Which number is in the 4 table *and* the 5 table *and* the 3 table?”